

Saddling The Tiger:
Trends & Critical Mechanisms
in Nuclear Weapons Acquisition

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Introduction:

The technology we wield shapes the wars that we fight. In their pioneering work on arms control, Thomas Schelling and Morton Halperin observed that certain weapons have inherent characteristics that can make military escalation more or less advantageous.¹ Especially vulnerable weapons may create pressures to use them before they are destroyed. Advocates of disarmament, arms control, and minimum deterrence—limiting the number of nuclear weapons to only those absolutely essential for a secure second strike—have elaborated on this basic argument: Nuclear weapons stationed alongside conventional weapons may result in unintended escalation, as an adversary may intend to destroy only conventional forces but instead bring attacks on nuclear forces into the window of acceptable action. Nuclear weapons that are indistinguishable from conventional weapons—placed on the same delivery vehicle, for instance—can risk ‘discrimination’ issues, where an adversary interprets a conventional attack as nuclear and responds accordingly.² Lower-yield nuclear weapons may reduce the costs of nuclear employment, correspondingly raising the risk of nuclear war.

Many of these points are debated. Would a state really panic and initiate full-scale nuclear war in the face of a few of their many nuclear weapons being destroyed? With modern sensors, would a state not observe the effects of a low-yield or conventional weapon before deciding how to respond, avoiding the discrimination problem?³ And if an adversary is already deploying low-yield nuclear weapons, would deploying a corresponding force not help convince the adversary that tactical nuclear war would be ineffective, and should not be pursued? Many advocates of nuclear deterrence and diverse nuclear arsenals believe the answer to all of these questions is ‘no’—that weapons do not create risk in these ways. However, they still believe that the technology we wield shapes the wars that we fight—or rather the wars that we avoid. They believe the weapons we deploy shape adversary perceptions and influence adversary decisions to go to war. They believe that adversaries may ponder the balance of nuclear forces when considering escalation, and that a diverse and large nuclear arsenal may be critical to deterrence.

In this manner, a majority of the nuclear policy community believe that decisions to integrate nuclear hardware—the weapons we develop and deploy—are important. It is therefore important to

¹ Thomas C. Schelling and Morton H. Halperin, *Strategy and Arms Control* (New York: Twentieth Century Fund, 1961), 9.

² James M. Acton, “Why Is Nuclear Entanglement So Dangerous?,” Carnegie Endowment for International Peace, January 23, 2019, <https://carnegieendowment.org/posts/2019/01/why-is-nuclear-entanglement-so-dangerous>.

³ Austin Long, “Discrimination Details Matter: Clarifying an Argument About Low-Yield Nuclear Warheads,” *War on the Rocks*, February 16, 2018, <https://warontherocks.com/discrimination-details-matter-clarifying-argument-low-yield-nuclear-warheads/>.

understand why states develop the particular nuclear weapons that they do. Many authors have written on this subject, but few have paid explicit attention to the justifications for nuclear diversification, and fewer still have done so through the means of detailed historical case studies. Those that have done so focus on deterrence more broadly as a category, including nuclear testing, doctrine, and weapons development in their studies, or on specific categories or aspects of nuclear weapons, such as strategic bombers or missile guidance.

This study examines two radically different cases of nuclear weapons acquisition—one long-past, and one ongoing—with a particular focus on the key actors involved in the acquisition process and the means by which decision-makers exert influence on the outcome of the case. In doing so, this research attempts to identify broad characteristics of nuclear weapons acquisitions as a category, as well as specific paths of influence and details of the acquisition process that have gone previously unnoticed. It identifies the congressional-executive-military relationship as a key venue of contestation, where specific strategic arguments are leveraged by the armed forces to gain congressional support in the face of executive opposition. More broadly, it presents a generalized theory of nuclear weapons acquisition where different parties leverage conditional advantages to achieve their goals.

Section I: Literature Review

Weapons, Holistically

Why do states build the weapons that they do? This question lies at the core of any research on weapons system adoption. States are complex organisms shaped by the interactions of many different internal actors. Different writers have different theories about how those actors come together to shape decisions about weapons acquisition. Adam Grissom, reviewing the field in 2006, provided a neat typology of these differing opinions.⁴ He grouped the field into schools, each named after the actor they believe is most prominent in the process of weapons acquisition—often called ‘military innovation’. Grissom found four schools:

- **Civil-Military Relations:** This school believes that political actors—the President and his appointees mainly, but also Congress—are the primary engine driving military innovation, but

⁴ Adam Grissom, "The Future of Military Innovation Studies," *Journal of Strategic Studies* 29, no. 5 (2006): 905–34.

only become involved in the process at certain junctures, such as military catastrophe or when domestic politics provide sufficient motivations for them to do so.⁵

- Inter-service Relations: This school believes that military innovation is primarily driven by competition between services—the U.S. Air Force, Navy, and Army, primarily—for resources in a zero-sum bureaucratic fight, where innovation serves to justify budget increases (Allison & Morris, 1975) (Bacevitch, 1986).
- Intra-service Relations: This school believes that relations between higher-ups *within* the services govern what military innovations get adopted—military leaders must successfully articulate a ‘theory of victory’ for the new innovation, and establish pathways to power for other military actors who agree with them, in order to push through an innovation.⁶
- Strategic Culture: This school believes that military innovation is controlled by the culture of the military unit—its beliefs, behavior, experience, and narratives. Some believe that this strategic culture is well-entwined with national identity, and changes little —others believe strategic culture can be manipulated by savvy military or political actors.⁷

These four schools are good for understanding the broad trends in the field of military innovation studies. However, they neglect to mention the influence of actors who are not military or government affiliated, but rather a part of defense contractors or other organizations with a stake in the military acquisition process. Additionally, they fail to represent the nuance of social interaction—all of these actors and drivers interact with one another in different ways at different times. No one party has complete control over military innovation—many of these variables, such as agreement within a service on a specific military innovation, are necessary but not sufficient for military innovation to occur.

Some theorists confront this problem by placing military innovation on a timeline, where different actors matter at different stages. Matthew Evangelista describes how the initial diffusion of an innovation occurs in military and scientific/technological communities, where strong networks of influence are particularly critical. Later, the support of congresspeople and the public becomes necessary, and external threats by other states are required to convince these actors that a new

⁵ Barry R. Posen, *The Sources of Military Doctrine: France, Britain, and Germany between the World Wars* (Ithaca, NY: Cornell University Press, 1984), Deborah D. Avant, "The Institutional Sources of Military Doctrine: Hegemons in Peripheral Wars," *International Studies Quarterly* 37, no. 4 (December 1993): 409–30.

⁶ Stephen Peter Rosen, *Winning the Next War: Innovation and the Modern Military* (Ithaca, NY: Cornell University Press, 1991).

⁷ Dima Adamsky, *The Culture of Military Innovation: The Impact of Cultural Factors on the Revolution in Military Affairs in Russia, the US, and Israel* (Stanford, CA: Stanford University Press, 2010). Theo Farrell and Terry Terriff, eds., *The Sources of Military Change: Culture, Politics, Technology* (Boulder, CO: Lynne Rienner, 2002).

innovation is necessary. Finally, other factors like cost-effectiveness, desirability to allies, or importance as a bargaining chip in international negotiations influence the ultimate fate of the new technology.⁸ Theo Farrell's early work draws a similar conclusion: institutional politics matters early on in the innovation process, but later political considerations determine the outcome of innovation competition.⁹

Michael Horowitz and Shira Pindyck also designate phases of the military innovation process, dividing it into an even three: "invention, when new technologies or tactics are created... incubation... when the invention gains status and influence through both top-down and bottom-up processes... [and] implementation[,] where political will leads the community of interest to adopt the innovation." Each phase is subject to different conditions: invention requires consistent R&D funding, incubation the leveraging of social influence in what Horowitz and Pindyck describe as "top-down and bottom-up processes," and implementation draws on the civil-military relations school in describing the necessity of political intervention.¹⁰

Beyond these theorists that break innovation into phases, there are also those that identify other conditions that are likely to influence when different actors succeed. Mary Kaldor lays out an essential theory of military innovation: that it occurs within the bounds of military conservatism. Kaldor sets out to explain the consistency of the 'follow-on imperative' described by James Kurth, where militaries are motivated to constantly acquire new weapons systems that fill the same niche as previous ones, rather than develop totally innovative systems.¹¹

Kaldor dubs this process 'Baroque' weapons development, where systems gain improvements in metrics like speed and range, but not in their mission or role, which is governed by previously established institutional missions. She explains that baroque arsenals result when a defense industry is sovereign—responsible for accruing funding to itself—but dependent—in need of government contracts to survive. These conditions cause defense contractors to innovate in an attempt to gain a

⁸ Matthew Evangelista, *Innovation and the Arms Race: How the United States and the Soviet Union Develop New Military Technologies* (Ithaca, NY: Cornell University Press, 1988), 00.

⁹ Theo Farrell, *Weapons Without a Cause: The Politics of Weapons Acquisition in the United States* (London: Macmillan, 1997), 00.

¹⁰ Michael C. Horowitz and Shira Pindyck, "What Is a Military Innovation and Why It Matters," *Journal of Strategic Studies* 46, no. 1 (2023): 85–114.

¹¹ James R. Kurth, "The Political Economy of Weapons Procurement: The Follow-on Imperative," *American Economic Review* 62, no. 2 (May 1972): 304–11.

competitive edge over other companies, but only innovate within the bounds set by conservative military requirements for incrementally-better weapons systems.¹²

Kaldor's theory is helpful, but doesn't explain edge cases where radical innovation did occur, shaking up existing missions, such as the introduction of missiles or drones into the Air Force. Donald MacKenzie and Graham Spinardi, theorists of the Social Shaping of Technology (SST), depict military technology as a product of free-flowing social negotiation between many different actors, including previously-discussed actors such as politicians and the military, but also industry representatives and technologists.¹³ Spinardi in particular highlights the role of technologists in explaining the cases of radical innovation that Kaldor does not include in her model.¹⁴ Kimberly Marten provides a similar perspective, calling typical innovation governed by Kaldor's baroque model "constrained", whereas innovation that breaks out of this framework is "radical." Marten argues that radical change is caused by increased interactions between different policy communities, such as militaries and think tanks, introducing new ideas to each other and deviating from typical bureaucratic conservatism.¹⁵

Marten and Spinardi both provide insight into the dominant behavior that governs military innovation. Actors have existing interests—for example, military services, or at least some senior elements of them, are generally conservative and prefer conforming to existing missions. These interests find themselves in competition with other pressures, usually stemming from other actors with opposed interests—for example, technologists, who are incentivized to sell new innovations to the military. None of these actors are necessarily at the helm of the whole military innovation process, but each has strategies accessible to them that can improve their chances of success, and might have more leverage at different phases of the process.

¹² Mary Kaldor, "The Weapons Succession Process," *World Politics* 38, no. 4 (July 1986): 577–95.

¹³ Donald MacKenzie, *Inventing Accuracy: A Historical Sociology of Nuclear Missile Guidance* (Cambridge, MA: MIT Press, 1990), 00.

¹⁴ Graham Spinardi, *From Polaris to Trident: The Development of US Fleet Ballistic Missile Technology* (Cambridge: Cambridge University Press, 1994), 00.

¹⁵ Kimberly Marten Zisk, *Engaging the Enemy: Organization Theory and Soviet Military Innovation, 1955–1991* (Princeton, NJ: Princeton University Press, 1993), 00.

Nuclear Weapons

A different set of scholarship exists that attempts to determine why states build nuclear weapons specifically, and then, once they've acquired nuclear weapons, why they acquire new nuclear delivery capabilities.

Scott Sagan's seminal article on the causes of nuclear proliferation identifies three models for why states build nuclear weapons: due to security pressures, due to domestic political forces, or due to normative incentives in the international system.¹⁶ Sagan's three-model thesis, in the same fashion as Grissom's typology of military innovation studies, oversimplifies nuclear proliferation. Steven Flank presents a thorough complication of Sagan's model based on SST theories. He argues that proliferation should be modeled as a complex interaction of social forces:

Nuclear weapons advocates need to recruit an array of allies... Nuclear system builders face limits on all resources—money, political authority and consensus, laboratory quality reagents, access to imports, and so on. The process by which these scarce resources are recruited and fixed in a stable network capable of producing the comparatively simple artifacts of 'nuclear weapons' *is the process of nuclear proliferation*.¹⁷

In Flank's framing, rather than simple conditions for nuclear proliferation, security pressures, bureaucratic forces, or normative incentives instead represent various actors within the social process of proliferation, or external conditions which they leverage to support their goal of nuclear weapons development. In this manner, the theory of nuclear proliferation is brought into line with the theory of military innovation more broadly.

Other scholars are interested in why states build different nuclear delivery capabilities, after acquiring nuclear weapons. Gartzke et. al identifies diversification of a nuclear arsenal broadly as a variable worth explaining, broadly concludes that it is a goal that all nuclear states seek, and offers explanatory variables for why states are more or less likely to achieve that goal. They identify experience with nuclear weapons and the nuclear capabilities of rivals and allies as key variables that contribute to

¹⁶ Scott D. Sagan, "Why Do States Build Nuclear Weapons? Three Models in Search of a Bomb," *International Security* 21, no. 3 (Winter 1996/97): 54–86.

¹⁷ Steven Flank, "Exploding the Black Box: The Historical Sociology of Nuclear Proliferation," *Security Studies* 3, no. 2 (Winter 1993/94): 259–94.

verification, and resource constraints and conventional defense requirements as constraining elements. Critically, the authors identify ‘bureaucratic politics’ as a possible explanatory variable for diversification, but quantify it by personnel assigned to specific organizations, rather than qualitative assessment of their power or influence.¹⁸

Quantitative observation often fails to properly account for the complex power bureaucratic actors wield. Other scholars grow closer to adequately documenting this power, though their work, in recognition of the complexity of the social system of military acquisitions, often avoids claims to general relevance outside of its immediate context. Edmund Beard, in *Developing the ICBM*, lays out in exhaustive detail various bureaucratic battles over the first intercontinental ballistic missile, describing how a well-connected political appointee, Trevor Gardner, leveraged his alliance with a well-placed military man, Bernard Schriever, and well-known scientist John von Neumann brought the weapon into being.¹⁹ Harvey Sapolsky, in a similar work on the first U.S. Submarine-Launched Ballistic Missile, also explores the bureaucratic struggle that led to the program’s completion, identifying specific strategies for bureaucratic success, including co-opting adversaries, moderating radical requirements, and creating innovative managerial procedures.²⁰ Donald Mackenzie and Graham Spinardi—previously mentioned in the section on military innovation—both claim to provide broad insights regarding the process of military technology formation, but focus specifically on nuclear weapons in their case studies—Mackenzie on accuracy in nuclear-tipped missiles, and Spinardi on the overall evolution of submarine-launched ballistic missiles, meaning their broader insights regarding the leverage of technologists in the process, and the social nature of weapons development, can be leveraged in a nuclear context.²¹

While these authors all examine specific weapons systems in close historical detail, some scholars work across a broad swathe of nuclear weapons development. *Flying Blind: The Politics of the U.S. Strategic Bomber Program* by Michael Brown examines the entire history of U.S. strategic bomber development, finding that neither technological innovation nor the economic interests of defense contractors are the impetus for new weapons development. Instead, the military services, sometimes working out of

¹⁸ Erik Gartzke, Jeffrey M. Kaplow, and Rupal N. Mehta, "The Determinants of Nuclear Force Structure," *Journal of Conflict Resolution* 58, no. 3 (2014): 481–508.

¹⁹ Edmund Beard, *Developing the ICBM: A Study in Bureaucratic Politics* (New York: Columbia University Press, 1976), 00.

²⁰ Harvey M. Sapolsky, *The Polaris System Development: Bureaucratic and Programmatic Success in Government* (Cambridge, MA: Harvard University Press, 1972), 00.

²¹ Mackenzie, *Inventing Accuracy*. Spinardi, *From Polaris to Trident*.

strategic necessity and sometimes out of parochial interest, are the primary pushers for new weapons.²² Janne E. Nolan identifies the mechanism by which the military exerts such control, describing, in her book *Guardians of the Arsenal*, the long history of ‘guardianship’ by the military over nuclear issues. By controlling war planning and target selection, military organizations enjoy a monopoly over expertise regarding what weapons are necessary to carry out their plans.²³

James Lindsay differs from Nolan and Brown, arguing that congress plays a significant role in nuclear weapons acquisition. He claims that after the Vietnam War and Watergate decreased congressional trust in—and therefore deference to—the executive, it became more active in nuclear decisionmaking. He also argues that Congress, rather than being motivated to intervene in weapons acquisitions by the prospect of accruing defense dollars for their district or decrying poor management of U.S. taxpayer funds, largely acts on issues of strategy and public policy.²⁴

Theory of Acquisition

In the same manner as military innovation holistically, nuclear arsenal development and the adoption of defense missions are ultimately social battles, where interested actors leverage strategies, and exploit conditional advantages, to fight for their preferred outcome. The military enjoys key advantages in this fight, conferred by the principle of guardianship, but is not always homogenous in its opinions. And when political actors can muster diverse networks of support such as in the case of ICBM development, they can overcome military interests. Additionally, specific points in the process—such as decisions to move into full production—are susceptible to congressional and therefore public influence. Meanwhile, technologists enjoy epistemological dominance over the realm of technical possibility, and can use it to constrain outcomes in accordance to their preferences.

The following section utilizes two case studies to loosely test the assertions of this theory, and identify other key factors in nuclear weapons acquisition. The two case studies were selected primarily due to their radical differences: the sea-launched nuclear cruise missile (SLCM-N) is a naval missile being developed in the present security environment; the B-70 bomber was an air force strategic bomber proposed in the late 1950’s and early 1960’s, during the Cold War. Similarities between these two cases should be rare—if they arise, they could serve as generalizable aspects of nuclear weapons development.

²² Michael E. Brown, *Flying Blind: The Politics of the U.S. Strategic Bomber Program* (Ithaca, NY: Cornell University Press, 1992), 00.

²³ Janne E. Nolan, *Guardians of the Arsenal: The Politics of Nuclear Strategy* (New York: Basic Books, 1989).

²⁴ James M. Lindsay, *Congress and Nuclear Weapons* (Baltimore: Johns Hopkins University Press, 1991), 00.

Section II: SLCM-N

In 2018, the Trump Administration published its first nuclear posture review. Citing Russia's belief in the strategic advantages of tactical nuclear weapons, the review describes the need for the United States to develop flexible and responsive low-yield nuclear options to "ensure that potential adversaries perceive no possible advantage in limited nuclear escalation..." One such option the review suggests is the sea-launched nuclear cruise missile, or SLCM-N.²⁵ Replacing a similar capability withdrawn in 1992 and retired in 2013, the SLCM-N would be based on attack submarines and surface ships, providing an option for limited nuclear employment that would bypass the long generation times required to deploy aircraft to a regional flare-up.

Its supporters claim that this capability is necessary to discourage the first use of tactical nuclear weapons by an adversary. The past three commanders of STRATCOM have supported the missile's development for this reason²⁶, accompanied by several republican members of congress and authors at think tanks such as the American Enterprise Institute and the Atlantic Council.²⁷ However, in 2022, the Biden administration's nuclear posture review called for the cancellation of SLCM-N, assessing that low-yield deterrence was adequately covered by the W76-2, a low-yield warhead that could be launched from existing ballistic missile submarines. The posture review also cited cost and lack of leveragability for arms control negotiations as additional reasons for the missile's cancellation.

The Biden Administration's attempt to cancel the program would prove unsuccessful. Weeks after the release of the classified Nuclear Posture Review, Members of the Nuclear Weapons Council—the

²⁵ Office of the Secretary of Defense, *Nuclear Posture Review* (Washington, DC: Department of Defense, February 2018), XI-XII.

²⁶ John R. Harvey and Robert Soofer, *Strengthening Deterrence with SLCM-N* (Washington, DC: Atlantic Council, 2022), <https://www.jstor.org/stable/resrep44945>.

U.S. Congress, Senate, Committee on Armed Services, Subcommittee on Strategic Forces, *To Receive Testimony on United States Strategic Command and United States Space Command in Review of the Defense Authorization Request for Fiscal Year 2026 and the Future Years Defense Program*, 119th Cong., 1st sess., March 26, 2025, transcript, <https://www.stratcom.mil/Media/Speeches/Article/4137505/sasc-fiscal-year-2026-us-strategic-command-and-us-space-command-posture-hearing/>.

U.S. Congress, House, Committee on Armed Services, Subcommittee on Strategic Forces, *Fiscal Year 2027 Strategic Forces Posture Hearing*, 119th Cong., 2nd sess., March 17, 2026, transcript, <https://www.stratcom.mil/Media/Speeches/Article/4437599/hasc-sf-fiscal-year-2027-strategic-forces-posture-hearing/>.

²⁷ Kyle Balzer, "SLCM-N and the Deterrence Value of Ambiguity," *Breaking Defense*, February 28, 2024, <https://www.aei.org/commentary/slcm-n-and-the-deterrence-value-of-ambiguity/>, Harvey & Soofer, *Strengthening Deterrence*. O'Brien, Connor, "Top Military Leaders Split with Biden over Nuke Cruise Missile," *Politico*, April 5, 2022, <https://www.politico.com/news/2022/04/05/nuke-cruise-missile-00023034>.

broad executive group responsible for the co-ordination of nuclear weapons—would testify before the Senate Armed Services Committee and claim the SLCM-N was now necessary to uphold deterrence. Admiral Charles Richard, commander of STRATCOM, and Admiral Christopher Grady, Vice Chairman of the Joint Chiefs of Staff, were most prominent in their support for the missile, with Admiral Grady claiming “[alignment] with the Chairman [of the Joint Chiefs]” regarding SLCM-N.²⁸ The Chairman, General Mark Milley, would later express this himself, stating before the Senate Armed Services Committee that his opinion on the missile “didn’t change” after an earlier expression of support.²⁹ After receiving this testimony, the democratically controlled Senate voted to continue funding the program at a limited level for fiscal year 2023.

A key factor in the congressional decision to continue funding the program seems to have been Russia’s invasion of Ukraine. The classified version of the nuclear posture review recommending SLCM-N’s cancellation had been composed prior to, and was released mere weeks after, the invasion began. Many of the individuals who had been involved in the creation of the anti-SLCM nuclear posture review now came out in favor of the weapon, including the military leaders previously mentioned. When pressed by Democratic Senator Elizabeth Warren regarding his participation in the posture review and subsequent opposition to SLCM’s cancellation, Admiral Richards pointed out that “Ukraine and the crisis that we’re in happened after the Nuclear Posture Review” to defend his seeming deviation from the administration’s position.³⁰ It is likely that this event influenced other military leaders in the administration similarly, and is at least partially responsible for the outpouring of support that military officials gave SLCM-N in their congressional testimony.

The chairman of the Senate Armed Services Committee’s Strategic Forces subcommittee, Senator Angus King, also seemingly favored the development of the missile. While the Senator is an

²⁸ U.S. Congress, Senate, Committee on Armed Services, Subcommittee on Strategic Forces, *Hearing to Receive Testimony on the Nuclear Weapons Council*, 117th Cong., 2nd sess., May 4, 2022, transcript, <https://www.stratcom.mil/Media/Speeches/Article/3022885/senate-armed-services-committee-hearing-nuclear-weapons-council/>.

²⁹ U.S. Congress, Senate, Committee on Armed Services, *Hearing to Receive Testimony on the Department of Defense Budget Posture in Review of the Defense Authorization Request for Fiscal Year 2023 and the Future Years Defense Program*, 117th Cong., 2nd sess., April 7, 2022, transcript, 00, https://www.armed-services.senate.gov/imo/media/doc/22-26_04-07-2022.pdf.

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<https://www.stratcom.mil/Media/Speeches/Article/3022885/senate-armed-services-committee-hearing-nuclear-weapons-council/>

independent and caucuses with the Democratic party, he nevertheless criticized the posture review in a hearing, pointing out that low-yield capabilities such as SLCM would fill a current “gap in our deterrent capacity.”³¹ Despite their ostensible majority, the overall position of the democratic party in the Senate at this time was weak. The senate was split 50-50, with two Democrat-affiliated independents filling out the blue side of the aisle. Vice President Kamala Harris frequently had to serve as a tie-breaker to push democratic legislation through. The relative dependence of the party on Senators like Angus King may have predisposed the committee to rule in favor of his pro-SLCM-N position, funding the missile to a limited extent rather than pushing hard for a full cancellation.

The position of congress would continue to turn in SLCM-N’s favor. In September of 2022, a slate of nuclear experts would be called to testify before the full Senate Armed Services Committee. Franklin C. Miller, Deputy Assistant Secretary of Defense for Nuclear Forces and Arms Control Policy under President George H.W. Bush, along with Ambassador Eric Edelman, former Undersecretary of Defense for Policy, jointly testified that congress should “restore funding for [SLCM-N] whose utility has been attested to by both Chairman Milley and Admiral Richard.” Rose Gottemoeller, former Under Secretary of State for Arms Control and International Security, and Madelyn Creedon, former Principal Deputy Administrator of the National Nuclear Security Administration spoke in opposition to the program, addressing the operational burden SLCM-N could place on naval attack submarines and the extent to which the long-range standoff air launched cruise missile could serve as a flexible replacement for SLCM-N. However, Miller scored a convincing last word on the matter, asserting that a limited program could be justified regardless of operational difficulties, as the finished missile is “in the out years,” and the Navy would have plenty of time to adapt while studies and development progressed.³²

The November 2022 congressional elections improved SLCM-N’s chances at continued funding as well, as the Democrats lost their grip on the House of Representatives. Under the new congress, testimony in favor of SLCM-N would continue, with Anthony Cotton, the new commander of STRATCOM, speaking favorably about the system in March 2023 hearings regarding the FY2024 budget.³³ Civilian experts would also continue to support the system: Gregory Weaver, a former

³¹Ibid

³²U.S. Congress, Senate, Committee on Armed Services, *United States Nuclear Strategy and Policy*, 117th Cong., 2nd sess., September 20, 2022, S. Hrg. 117–962.

³³U.S. Congress, Senate, Committee on Armed Services, *To Receive Testimony on United States Strategic Command and United States Space Command in Review of the Defense Authorization Request for Fiscal Year 2024 and the Future Years Defense Program*, 118th Cong., 1st sess., March 9, 2023, transcript,

nuclear official on both the Joint Staff and the Office of the Under Secretary of Defense for Policy, Dr. Brad Roberts, director of the Lawrence Livermore National Laboratory Center for Global Security Research, and Elaine Bunn, former deputy assistant Secretary of Defense for nuclear and missile defense policy, would all testify in favor of the program in an additional March hearing before the Senate Armed Services Committee Strategic Forces subcommittee. In the same hearing, Senator King would continue to press for the weapon's development, stating that he was "puzzled why [the program] is not in the budget" considering the significant testimony his subcommittee had received in favor of it.³⁴ In a later hearing, King would pressure Vice Admiral Johnny Ray Wolfe, naval director for Strategic Systems Programs, about the radical change since the nuclear posture review, especially Russian "discussion of the use of tactical nuclear weapons", and expressed hope that the administration would "continue to review that issue."³⁵ While some administration officials such as Secretary of Energy Jennifer Granholm stuck by the decision to cancel SLCM-N, they typically did so by referring to the Nuclear Posture Review as their guidance, rather than expressing justifications for the missile's cancellation.³⁶

The SASC subsequently recommended a FY2024 provision that would require the executive branch to begin developing the missile and its accompanying warhead. But before the year's defense budget could be passed, a new panel of experts would weigh in on SLCM-N. The Strategic Posture Commission was a group of nuclear experts, established by congress in FY2022 and designed to provide a reasoned, third party assessment of the threat environment that the nation faced, and the appropriate nuclear weapons policy to confront that threat. The commission released its report in October of 2023. It included a fully-throated statement of support for a theater-tailored system that would be "forward-deployable...

<https://www.stratcom.mil/Media/Speeches/Article/3326284/sasc-fiscal-year-2024-us-strategic-command-and-us-space-command-posture-hearing/>.

³⁴U.S. Congress, Senate, Committee on Armed Services, *Hearing to Receive Testimony on the Department of Defense Budget Request for Fiscal Year 2024 and the Future Years Defense Program*, 118th Cong., 1st sess., March 28, 2023, transcript, https://www.armed-services.senate.gov/imo/media/doc/23-21_03-28-2023.pdf.

³⁵ U.S. Congress, Senate, Committee on Armed Services, Subcommittee on Strategic Forces, *Hearing to Receive Testimony on the Department of Energy's Atomic Energy Defense Activities and Department of Defense Nuclear Weapons Programs in Review of the Defense Authorization Request for Fiscal Year 2024 and the Future Years Defense Program*, 118th Cong., 1st sess., April 18, 2023, transcript, https://www.armed-services.senate.gov/imo/media/doc/23-28_04-18-2023.pdf.

³⁶ U.S. Congress, Senate, Committee on Armed Services, *Hearing to Receive Testimony on the Department of Energy and National Nuclear Security Administration Atomic Energy Defense Activities in Review of the Defense Authorization Request for Fiscal Year 2024 and the Future Years Defense Program*, 118th Cong., 1st sess., April 26, 2023, transcript, https://www.armed-services.senate.gov/imo/media/doc/23-35_04-26-2023.pdf.

survivable against preemptive attack without force generation” and allow for “a range of explosive yield options, including low yield.”³⁷ While the commission’s report did not explicitly name SLCM-N, the requirements it laid out could be effectively fulfilled only by the program. The Strategic Forces Subcommittee quickly held hearings on the subject, where Senator King, reflecting on the commission’s findings, again stated that the “threat of Russians using tactical weapons in Ukraine has sort of woken us up... [and] brings us back to SLCM-N”, a statement quickly supported by former Senator Jon Kyl, vice chair of the commission.³⁸ Additionally, Madelyn Creedan, who had previously given testimony indicating skepticism toward the operational challenges presented by SLCM-N, was the chair of the strategic posture commission, indicating a broader support for the system than had previously been evident.

At the end of the year, the congress passed the National Defense Authorization Act requiring the Secretary of Defense to set up development work for SLCM-N, despite the Biden Administration’s public opposition.³⁹ Congress has continued to vote for consistent levels of funding for the weapons system in subsequent years, leading the program to steadily progress. The Undersecretary of Defense for Acquisitions and Sustainment testified in March of 2024 that he had directed the Navy to establish a program office for the missile. On August 22nd of 2025, the Strategic Systems Programs division of the US Navy awarded prototype design contracts to five contractors: Leidos, Raytheon, Lockheed Martin, and Northrop Grumman were tasked with flight system integration and missile development, while Florida Turbine Technologies was made responsible for missile technologies development. The contract awards followed a two-day Industry Day managed by the National Security Technology Accelerator, where non-traditional defense contractors and traditional primes were brought into contact with navy officials and could coordinate on the direction for the missile. The prime contractors selected will, according to the Accelerator, lead teams of “traditional and non-traditional defense contractors that will spearhead the initial phase of SLCM-N flight system design.”⁴⁰

³⁷ Congressional Commission on the Strategic Posture of the United States, *America’s Strategic Posture: The Final Report of the Congressional Commission on the Strategic Posture of the United States* (Washington, DC: Institute for Defense Analyses, October 2023), 100, <https://fas.org/wp-content/uploads/2023/10/Strategic-Posture-Committee-Report-Final.pdf>.

³⁸ “King Highlights Importance of Nuclear Deterrence to Better Protect Against Adversaries,” press release, Office of Senator Angus King, October 19, 2023, <https://www.king.senate.gov/newsroom/press-releases/king-highlights-importance-of-nuclear-deterrence-to-better-protect-against-adversaries>.

³⁹ Executive Office of the President, Office of Management and Budget, “Statement of Administration Policy: S. 2226 – National Defense Authorization Act for Fiscal Year 2024,” July 27, 2023, <https://www.whitehouse.gov/wp-content/uploads/2023/07/S2226-NDAA-SAP-Followon.pdf>.

⁴⁰ “U.S. Navy Awards SLCM-N Prototype Effort to Advance Sea-Based Nuclear Deterrent,” NSTXL, September 3, 2025, <https://nstxl.org/us-navy-awards-slcmn-prototype/>.

In the November 2024 election, the Republican party gained control over the White House and Senate, and retained its majority in the House of Representatives. The second Trump Administration quickly reversed the Biden administration's policy on SLCM-N, re-including it in presidential budget requests. Secretary of Defense Pete Hegseth stated his support for the program in his confirmation hearing.⁴¹ The administration and military leaders have henceforth expressed a unified front of support for the program. Assistant Secretary of War for Nuclear Deterrence, Chemical, and Biological Defense, Policy and Programs Dr. Robert Kadlec indicated his support for the program in a 2026 hearing before the House Armed Services Committee, as did STRATCOM commander Admiral Richard Correll.⁴² The atmosphere of SLCM-N development contract selection—accelerated, non-traditional, and agile—fits with both the overall ethos of the Trump Administration's acquisition reform, and the emerging military ambition for the program. Steady bi-cameral support for the program has ignited ambitions to speed it up: in 2025, Admiral Cotton, then STRATCOM commander, when questioned by the House Armed Services Committee about the tradeoff between “survivability versus time to field” for SLCM-N, emphasized the necessity of “see[ing the program] accelerate and not languish to the right”, and that he would “always love to see [the Navy] move faster” to acquire SLCM-N.⁴³ He would later repeat a similar sentiment in the Senate Armed Services Committee. While some Senators, such as Senator Mark Kelly, continued to express concerns regarding the operational constraints caused by SLCM-N, the committee largely continued to fund SLCM-N on a “bipartisan basis.”⁴⁴

Given the previous support for SLCM-N by a democratically-controlled Senate and House, it seems unlikely that congress will, in the foreseeable future, return to restricting funds for SLCM-N, even in the event that it returns to democratic control. It is possible that a democratic landslide in a coming congressional election could eliminate the party's hypothesized dependence on Senator Angus King, and its need to cave to the Senator's support for SLCM-N. However, regardless of King's relative power in a future congress, many democratic congresspeople have likely come to support the program as time has progressed. In fact, there is some indication that the Biden Administration had changed its position on SLCM-N by the end of its tenure. In August of 2024 Deputy Assistant Secretary of Defense for Space Policy Vipin Narang spoke at the Center for Strategic and International Studies,

⁴¹ Walter Pincus, “Pete Hegseth's Answers to Questions That Weren't Raised in Public,” *The Cipher Brief*, January 21, 2025, https://www.thecipherbrief.com/column_article/pete-hegseths-answers-to-questions-that-werent-raised-in-public.

⁴² U.S. Congress, *Fiscal Year 2027 Strategic Forces Posture Hearing*.

⁴³ U.S. Congress, *To Receive Testimony on USSTRATCOM and USSPACECOM in Review of Defense Authorization Request for Fiscal Year 2026*

⁴⁴ Ibid

indicating that the administration was not simply complying with the congressional mandate for SLCM-N, but had, in some respects, recognized the changes in the security environment that had caused congress to force funding through for the program:

“The 2022 NPR canceled the program because at the time the administration assessed it was unnecessary and could distract from other priorities. **Today the world is different.** Put clearly, the department is proceeding with a SLCM-N and we’re working closely with Congress to ensure we are meeting our shared goals of getting the most deterrence value for the least risk to the modernization program, the nuclear-weapons complex, and the joint force.”⁴⁵

The fact that a democratic administration, and officials working within it, had come, gradually, to favor SLCM-N, is further indication that a future democratic congress is unlikely to restrict funding for the program.

However, it is possible that a future democratic administration takes a similar posture to the Biden administration’s stance from 2023 to 2025—that while the salience of tactical nuclear weapons seems to be increasing, the U.S. should still avoid developing new systems such as SLCM-N. In the event that this hypothetical administration comes to power, it may find an unlikely ally in the Navy. The only firm statement from a Chief of Naval Operations on SLCM-N was made in 2022, and concerns operational difficulties with implementing the weapon. Admiral Michael Gilday stated, before the House Armed Services Committee, that he was “not convinced yet that we need to make a \$31 billion investment in that particular system to close that particular gap.” High-ranking naval officers have continued to raise similar concerns in more recent years: in late 2024, Vice Admiral Johnny Wolfe, director of Strategy Systems Programs, called the projected 2034 deployment date for SLCM-N a “very, very aggressive timeline” and cited the need for an “entire new workforce and industrial base.”⁴⁶ In May of 2025 he questioned whether the Navy would be able to, by 2034, “get this system integrated into a platform that was never purpose built for [it]”, referring to the Virginia-class submarine.⁴⁷ The

⁴⁵“Nuclear Threats and the Role of Allies: A Conversation with Acting Assistant Secretary Vipin Narang,” Center for Strategic and International Studies, August 1, 2024,

<https://www.csis.org/analysis/nuclear-threats-and-role-allies-conversation-acting-assistant-secretary-vipin-narang>.

⁴⁶Sydney J. Freedberg Jr., “Sub-Launched Nuclear Cruise Missile Will Need ‘an Entirely New Industrial Base,’ Warns Navy Admiral,” *Breaking Defense*, November 15, 2024, <https://breakingdefense.com/2024/11/sub-launched-nuclear-cruise-missile-will-need-an-entirely-new-industrial-base-warns-navy-admiral/>.

⁴⁷Valerie Insinna, “Integration on Virginia-Class Subs the ‘Greatest Risk’ for Nuclear Sea-Launched Cruise Missile: Navy Official,” *Breaking Defense*, May 21, 2025,

Navy's continued concerns for the viability of the program could serve as an effective lever for an incoming administration to justify its cancellation.

Analysis

While Congress was the most effective and critical actor in the debate over SLCM-N, the degree to which it acted on its own behalf is difficult to assess. Congress's responsiveness to public sentiment on this issue seems minimal—while some think tanks released articles in favor of SLCM-N, others released corresponding pieces against the weapons system. No substantial or widespread public action movement against SLCM-N existed at the time. Instead, it seems that congress largely responded to military advocacy for the weapon, with the House and Senate Armed Services committees serving, essentially, as representatives for the military.

In that manner, SLCM-N demonstrates the principle that guardianship confers an advantage to military actors in nuclear policy decisionmaking. However, it shows that only specific actors can convincingly leverage that advantage—in this case, STRATCOM and the Joint Chiefs of Staff. While STRATCOM's assertion regarding the necessity of nuclear flexibility was taken seriously by congress and led to further appropriations for SLCM-N, the Navy's corresponding expertise regarding the difficulty of implementing the missile into their existing forces was largely ignored. This shows that congress seems to be more willing to listen to strategic arguments than those that address costs or technical constraints, supporting James Lindsay's assertions in *Congress and Nuclear Weapons*.

The Biden administration could not affect congress's support for SLCM-N. Whether this was due to its inability of this specific administration to adequately build a coalition around opposition to the weapons system, or rather a more structural overarching difficulty in the ability of the executive to suppress nuclear weapons programs, is difficult to determine given the limited scope of this article. However, given only this case, the true extent of the power of the political wing of the executive branch seems limited in regards to nuclear weapons.

Additionally, technologists seem to have had little leverage thus far in SLCM-N's inception and initial development. There is no evidence that Trump's initial support for the weapon was inspired by influence from industry, and the congressional decisionmaking processes does not seem to have been

<https://breakingdefense.com/2025/05/integration-on-virginia-class-subs-the-greatest-risk-for-nuclear-sea-launched-cruise-missile-navy-official/>.

influenced by corporate actors. However, it is notable that SLCM-N is utilizing the Other Transaction Authority, a contracting tool designed to reduce regulation on, and correspondingly speed up, military acquisitions. This lack of regulatory oversight could provide technologists with a situational advantage in the future.

Section III: The B-70

SLCM-N is being driven along by a number of key actors: congress, civilian experts, the Trump administration, and leaders of STRATCOM and the Joint Chiefs. These diverse actors have come together to support the weapons system based on one key evaluative metric: the relative flexibility that it provides the President. In the nuclear context, flexibility can be defined as the range of options a political leader has to respond to a given scenario. For deterrence to be preserved, political leaders must have convincing options to respond to all levels of adversary escalation. If the leader has limited options—for instance, only being able to respond to a small-scale, low-yield nuclear use with an all-out attack on their adversaries' cities—then the credibility of their response is limited.

Flexibility is a longstanding concept in nuclear weapons policy, finding its origins in the late 1950s and early 1960s. During this period, flexibility was intimately tied to the interest of the Air Force in sustaining their manned bomber force. The particular bomber on which the fate of the force hung was the B-70, which was eventually cancelled by Secretary of Defense Robert McNamara. The following section explores the origins of flexibility through the lens of the fight over the B-70, while drawing other critical conclusions about civil-military relations in defense acquisitions.

In the mid 1950s, the Strategic Air Command (SAC)—the body within the Air Force responsible for the management of strategic nuclear weapons—was dissatisfied with their bomber arsenal. Studies by the RAND corporation had indicated that their upcoming supersonic bomber, the B-58, would be vulnerable to Soviet pre-emptive attacks. Additionally, the first Soviet intercontinental bomber was unveiled in May of 1954, sparking worries in the Air Force regarding its ability to match Soviet strategic capabilities.⁴⁸ In response, in October of the same year, SAC published extremely ambitious requirements for its next strategic bomber, a supersonic, high-altitude, long-range aircraft intended to push the state of the aviation art as far as it could go. The aircraft, dubbed WS (Weapons System) 110-A, also came with an equally optimistic projected deployment date in the early 1960s.⁴⁹

⁴⁸ Brown, *Flying Blind*, 198.

⁴⁹ Kenneth P. Katz, *The Supersonic BONE: A Development and Operational History of the B-1 Bomber* (Barnsley, UK: Pen & Sword Aviation, 2022), 7.

7, Brown, *Flying Blind*, 201-202.

The aircraft quickly ran into trouble: in a presentation to Air Force Headquarters, the Wright Air Development Center acknowledged that their projected schedule would prove tight. Two contractors for the system—Boeing and North American—were selected in December of 1955, but both companies' initial designs were based on massive, detachable fuel tanks that could be attached to equally gargantuan wings. SAC Commander Curtis LeMay derided the designs as "three plane formation[s]" and told the contractors to return to the drawing board.⁵⁰ Elsewhere however, the scientific frontier was beginning to meet SAC's ambition. In March of 1956 the National Advisory Committee on Aeronautics published a memorandum on 'compression lift', a means of capturing the shock waves from supersonic flight and using them to increase the lift of an aircraft. In late 1956, the WS-110 contractors came across this study, and North American began to incorporate it into its aircraft designs. While compression lift would require a number of radical innovations, including new materials, engines, inlets, weapons systems, and navigation, North American had met the Air Force's ambition, and was selected to build the WS-110.⁵¹ In the meantime, the Air Force's other new bomber, a proposed nuclear-powered aircraft, was cancelled, leaving the WS-110 the service's only shot at a manned aircraft for the future.⁵² In February of 1958, the WS-110 program was designated the B-70—the first flight of the advanced aircraft was expected only 35 months later, in December of 1961.⁵³

The aircraft would not proceed as planned. A low-altitude requirement was added in March of 1958 in response to the emerging threat of high-altitude Soviet air defense, placing additional demands on the plane's budget.⁵⁴ Later that year, as intercontinental missile development came to dominate the Air Force's budgetary priorities, the B-70 program was stretched to take up less budgetary space each year. This extended production schedule added 160 million to the total program costs, however.⁵⁵ In the fall of 1959, two associated programs intended to increase the capabilities and reduce the cost of the aircraft were cancelled—a high-energy fuel program intended to increase the aircraft's range, and the F-108 interceptor, which shared subsystem costs with the B-70. The F-108 cancellation drove the

⁵⁰ Katz, *The Supersonic BONE*, 7. Brown, *Flying Blind*, 206-208.

⁵¹ Brown, *Flying Blind*, 210-213, Gary Beatovich, "A Case Study of Manned Strategic Bomber Acquisition: The B-70 Valkyrie" (master's thesis, Air Force Institute of Technology, 1990), 25-27, <https://apps.dtic.mil/sti/tr/pdf/ADA229545.pdf>, Katz, *The Supersonic BONE*, 10.

⁵² Brown, *Flying Blind*, 209.

⁵³ Katz, *The Supersonic BONE*, 10. Beatovich, "A Case Study", 27.

⁵⁴ Brown, *Flying Blind*, 214-216.

⁵⁵ Brown, *Flying Blind*, 217.

B-70's costs up by another 180 million.⁵⁶ In December, the axe fell—the B-70 program was cut back to an exploratory prototyping program, without a set date for a development aircraft.⁵⁷

While cost overruns may have driven attention to the program's deficiencies, at its core, the cancellation was a result of ideological disagreement between President Eisenhower and the Air Force. In mid-November, just before the B-70's cancellation, Eisenhower met with key military and civilian national security advisors at Augusta National Golf Club to decide the fate of the program. Nick Kotz, journalist and author of a book on the B-70 and its successors, describes the proceedings: first the Chairman of the Joint Chiefs of Staff, Gen. Nathan F. Twining, advocated for the B-70's ability to serve as a reconnaissance-strike vehicle, a plane that could fly over the Soviet Union, hunt down, and destroy mobile missile launchers. Eisenhower found this suggestion ridiculous, and affirmed his belief that a nuclear war would be fought with massive, direct attacks on population centers, rather than with discriminate strikes on hidden missiles. Tommy White, Chief of Staff of the Air Force, then attempted to defend the B-70 on the grounds of its flexibility relative to missiles: it could be recalled, complicate Soviet defense plans, and provide a means of psychological communication due to the visibility of its deployment. Eisenhower rejected White's argument as well, stating that "we are talking about bows and arrows at the time of gunpowder when we speak of bombers in the missile age."⁵⁸

This discussion at Augusta was one of the first invocations of the concept of flexibility to defend manned bomber aircraft. While missiles could provide prompt assured destruction, bombers were more versatile, could allow for more tailored strikes, and could execute limited nuclear employment more effectively. Eisenhower rejected this idea, maintaining his adherence to the rigid conception of nuclear war known as massive retaliation. His administration believed that any war between the U.S. and the Soviet Union would escalate into a general war, and that any general war between the two powers would inevitably be nuclear in nature. While Air Force officials advocated for employment of nuclear weapons against military targets, namely enemy missiles, Eisenhower "could not conceive" of this strategy "as a feasible option."⁵⁹ In an inflexible massive nuclear conflict, flexible bombers would be necessary—hence his cancellation of the B-70.

⁵⁶ Brown, *Flying Blind*, 217-218. Bernard C. Nalty, *The Quest for an Advanced Manned Strategic Bomber: USAF Plans and Policies, 1961-1966* (Washington, DC: USAF Historical Division Liaison Office, 1966), 2, https://media.defense.gov/2026/May/04/2003923557/-1/-1/1/QUEST%20FOR%20AN%20ADVANCED%20MANDED%20STRATEGIC%20BOMBER_AFD-110321-046.PDF.

⁵⁷ Katz, *The Supersonic BONE* 11. Brown, *Flying Blind*, 218.

⁵⁸ Nick Kotz, *Wild Blue Yonder: Money, Politics, and the B-1 Bomber* (New York: Pantheon Books, 1988), 34-35, 57.

⁵⁹ David Alan Rosenberg, "The Origins of Overkill: Nuclear Weapons and American Strategy, 1945-1960," *International Security* 7, no. 4 (Spring 1983): 62-63.

However, while Eisenhower kept a firm grip on the Air Force's force structure, he was more loose with his control of military targeting practices. While he attempted to establish a joint nuclear targeting plan where each military service would have input, he inadvertently gave SAC free reign over targeting doctrine: the Joint Strategic Target Planning Staff was dominated by SAC officers due to the Navy's inability to, and the Army's lack of interest in, sending enough targeters to counter them.⁶⁰ The result was SIOP-62, the first Strategic Integrated Operational Plan for nuclear war, which included city targets but also a number of diverse military targets, mandating a massive nuclear force. Eisenhower had won the battle over the B-70 temporarily, but had ceded critical ground—the Air Force was now the key decider of how many and what kind of nuclear weapons it needed, and would leverage this position to keep fighting for its manned bombers.

In January of 1960, Tommy White announced that the Air Force would take that fight before congress—while Eisenhower contemplated firing the general for his break with the administration, he ultimately decided against it: the damage had been done.⁶¹ Senators from across the country—California, Arizona, Washington—began coming out against the administration's decision.⁶² While Eisenhower's Secretary of Defense, Thomas Gates, attempted to justify his decision in a congressional hearing by citing the “research stage” materials the system would require and its relative vulnerability to enemy air defense, the tide of B-70 support continued.⁶³ Former Air Force Secretary Missouri Senator Stuart Symington, alongside Texas Senator Lyndon Johnson, both came out in favor of the B-70.⁶⁴ Johnson went a step further and, as chairman of the Senate Armed Services Committee's Preparedness Subcommittee, helped author a committee report endorsing the plane. Meanwhile, Massachusetts Senator and Presidential candidate John F. Kennedy began advocating for the aircraft on the campaign trail.⁶⁵ In July of that year, congress voted in an additional \$265 million for the B-70 program—\$190 million more than the Eisenhower administration intended to spend.

While at first Eisenhower only added \$60 million to the program's budget, eventually his hand was forced.⁶⁶ To preserve the political position of his party against attacks from the Kennedy campaign and democratic senators, Eisenhower released \$20 million more in funds for B-70 development in fiscal year 1961, and reinstated its weapon system status: both prototype aircraft would now be equipped

⁶⁰ Ibid, 64-65.

⁶¹ Kotz, *Wild Blue Yonder*, 36.

⁶² Kotz, *Wild Blue Yonder*, 55

⁶³ Nalty, *Advanced Manned Strategic Bomber*, 3.

⁶⁴ Brown, *Flying Blind* 220.

⁶⁵ Kotz, *Wild Blue Yonder*, 62-63.

⁶⁶ Robert Frank Futrell, *Ideas, Concepts, Doctrine: Basic Thinking in the United States Air Force*, vol. 2, 1961–1984 (Maxwell Air Force Base, AL: Air University Press, 1989), 29. Brown, *Flying Blind*, 219-220

with weapons systems. Along with the B-70 funds, Eisenhower released \$456 million in other monies for defense spending.⁶⁷ He would quickly leverage this for explicitly political purposes, suggesting to a prominent business leader that, since the government had released money for defense spending, corresponding funds should be released by corporations to fund Vice President Richard Nixon's campaign.⁶⁸ And just days before the election, on October 31st, Eisenhower released an additional \$110 million for the B-70, announcing plans to fully develop the plane and eventually build 200 total aircraft. Kennedy's subsequent comments at a San Diego rally highlighted the fact that a majority of the work on the aircraft would be completed in California, a swing state, and questioned "why... they did it this week?"—calling out the Eisenhower administration's obviously political ploy. While Kennedy won the Presidential election, Richard Nixon pulled through in California.⁶⁹

The incoming Kennedy administration believed in the publicly-accepted theory of a 'missile gap'—that the Soviet Union had far more strategic nuclear missiles than the United States—and campaigned heavily on raising defense spending. However, upon taking office, the President's national security team quickly realized that the missile gap was, in fact, in the United States' favor—the US had far more strategic missiles than the Soviets. The Kennedy administration quickly adjusted their policy in accordance with this revelation—in March of 1961, they announced a reversal of Eisenhower's decision to fully develop the B-70. The weapons system would, once again, be limited to only experimental prototypes. Before congress, Kennedy's Secretary of Defense, Robert McNamara, would emphasize the sufficiency of the B-52 force for upholding strategic deterrence, and the B-70's inability to effectively penetrate Soviet air defenses.⁷⁰

Once again, the Air Force would fight to retain its next manned bomber. Throughout the Spring of 1961, Tommy White would appear before both the House Armed Services Committee and the House Appropriations subcommittee on Defense to advocate for the B-70, namely citing its flexibility and the necessity of systems that "can look, and find, and report, and attack, and return, and attack again."⁷¹ Meanwhile, McNamara initiated studies of possible B-70 alternatives, including an improved B-58 or a nuclear-powered aircraft.⁷² In July, Curtis LeMay, former commander of Strategic Air Command, was appointed as Air Force Chief of Staff to replace Tommy White. LeMay took up White's cause, and

⁶⁷ Futrell, *Ideas, Concepts, Doctrine*, 21-22. Brown, *Flying Blind*, 219.

⁶⁸ Kotz, *Wild Blue Yonder*, 62-63.

⁶⁹ Brown, *Flying Blind*, 219, Kotz, *Wild Blue Yonder*, 64, Nalty, *Advanced Manned Strategic Bomber*, 4.

⁷⁰ Brown, *Flying Blind*, 221-222, Futrell, *Ideas, Concepts, Doctrine*, 29-30

⁷¹ Futrell, *Ideas, Concepts, Doctrine*, 34. Zachary M. Matusheski, "Bomber Wars: Curtis LeMay, Robert McNamara and the B-70," *Air Power History* 58, no. 4 (Winter 2011): 19.

⁷² Nalty, *Advanced Manned Strategic Bomber*, 6.

resumed a vigorous campaign for the B-70, stating in a memo to McNamara sent only two days after his appointment that there were no “acceptable alternatives” to developing the plane.⁷³

LeMay and McNamara’s contest would decide the fate of the manned bomber for the next decade. The contrast between the two men is described best by Zachary Matusheski in his piece “Bomber Wars.” McNamara, a former CEO of the Ford Motor Company, brought with him to the pentagon a modern style of management based on technical, scientific, neutral analysis. His primary focus was on balancing capabilities and cost, achieving an effective deterrent force while spending as little money as possible. LeMay, meanwhile, had a long-time attachment to the use of the manned bomber in war, cultivated during the strategic bombing of Japan in World War II, and a strongly-held belief that the U.S. missile force could not achieve strategic deterrence on its own.⁷⁴

The two would hold their battle with the resources available to them: McNamara utilized his presidential access and executive control, while LeMay worked to influence congress. In a July memo to President Kennedy reviewing U.S. strategic forces, McNamara maintained his preference for hardened missiles over manned bombers.⁷⁵ Later that month, the military division of the bureau of the budget suggested putting a complete halt to current Air Force plane procurement.⁷⁶ Meanwhile, LeMay, joined by Secretary of the Air Force Eugene Zuckert, testified before the senate appropriations committee regarding the need to speed up B-70 development, arguing that a large portion of the technical issues that had troubled the program early on had been resolved.⁷⁷

LeMay’s congressional testimony would achieve his desired effect: the National Defense Authorization Act for fiscal year 1962 would include funds to turn the B-70 from a prototype into a new weapons system, and acquire a new wing of B-52s.⁷⁸ But the battle was by no means won. LeMay continued to speak across the country on the necessity of flexible manned bombers, and McNamara began working to override the decision of congress.⁷⁹ Prior to the passage of the 1973 Impoundments Control Act, the President could impound funds allocated by congress to specific programs, preventing them from being spent.. McNamara encouraged Kennedy to utilize this power to stop the B-70 funds from being given to the Air Force, highlighting the vulnerability of bombers and their lack of penetration

⁷³ Matusheski, “Bomber Wars”, 19.

⁷⁴ Ibid.

⁷⁵ Ian Clark, *Nuclear Diplomacy and the Special Relationship: Britain’s Deterrent and America, 1957–1962* (Oxford: Clarendon Press, 1994), 346-7.

⁷⁶ Matusheski, “Bomber Wars”, 19-20

⁷⁷ Matusheski, “Bomber Wars”, 19. Nalty, *Advanced Manned Strategic Bomber*, 4.

⁷⁸ Futrell, *Ideas, Concepts, Doctrine*, 34. Brown, *Flying Blind*, 224, Matusheski, “Bomber Wars”, 19.

⁷⁹ Ibid

capability.⁸⁰ On October 27th of 1961, McNamara announced the impoundment of the money allocated to the B-70 in August—congress immediately came out in opposition, with Senator John Stennis, chairman of the preparedness subcommittee, voicing his concerns to reporters that same day, and Senator Henry Jackson calling the decision “risky” soon after.⁸¹

McNamara would attempt to head off congressional opposition to his decision by restricting the ability of military leaders to speak to congress: he ordered in November that all communications be routed through his Assistant Secretary of Defense for Legislative Affairs.⁸² This strategy failed—LeMay and his Vice Chief of Staff Fredric H Smith Jr. would continue to testify before senate committees in January, and more Senators subsequently came out against McNamara’s impoundments in the New York Times.⁸³ Meanwhile, the Air Force was attempting to rework the B-70 concept into a plane explicitly tailored for flexibility—the Reconnaissance-Strike, or RS-70. The RS-70 would be equipped with advanced sensors to effectively carry out the ‘look-shoot-look’ mission that Tommy White had invoked. The Air Force began suggesting the reworked B-70 in October of 1961, submitting a proposal to McNamara which was met with significant skepticism.⁸⁴ In January, the service requested a release of \$80 million of the impounded B-70 funds to convert one B-70 prototype into an RS-70, estimating a need for a further \$320 million to fund the project through fiscal year 1963. McNamara accepted that the RS-70 could prove useful, but was still against a full-scale development program with an intent to eventually deploy the aircraft—rather, he favored a continuation of the prototype program with a preserved option to fully develop the aircraft at a later date. In February of 1962, the Air Force was formally informed that the RS-70 funding release had been denied due to skepticism about the ability of “the present state of the art... to support system development.” Shortly afterward, Secretary of the Air Force Eugene Zuckert testified against this delay in a House Armed Services Committee hearing.⁸⁵

The back-and-forth between congress and the executive would come to a head in March of 1962. Georgia Representative Carl Vinson, a Democrat dubbed the ‘Swamp Fox’ for his keen political ability, had long held with a strong grip the chairmanship of the House Armed Services Committee. Previously chairman of the House Committee on Naval Affairs before the creation of the Department of Defense, Vinson had developed a reputation as a commanding force in military matters. Upon President Kennedy’s election, Vinson and McNamara had a brief disagreement—McNamara was considering a plan to reorganize the Department of Defense and unify the armed services into a single

⁸⁰ Matusheski, “Bomber Wars”, 20.

⁸¹ Futrell, *Ideas, Concepts, Doctrine*, 35. Brown, *Flying Blind*, 224, Matusheski, “Bomber Wars”, 20.

⁸² Matusheski, “Bomber Wars”, 20.

⁸³ Ibid, 20-21

⁸⁴ Futrell, *Ideas, Concepts, Doctrine*, 51, Brown, *Flying Blind*, 223

⁸⁵ Nalty, *Advanced Manned Strategic Bomber 7*. Futrell, *Ideas, Concepts, Doctrine*, 51-52.

service. Vinson deeply opposed the plan, and said so, along with Senator Richard Russel of the Senate Armed Services Committee, to McNamara—McNamara subsequently dropped the plan, and Vinson hailed him as a potential “best Defense Secretary the country has had.”⁸⁶

While Vinson and McNamara always expressed public admiration for each other—in November of 1961 McNamara called himself “the newest pupil in the Russell-Vinson school for secretaries of defense,”⁸⁷—the brief dispute over DoD organization would not be their only battle. In February of 1961, Vinson extracted a compromise from the Secretary—he would support the Kennedy Administration’s budget only if McNamara promised to make manned bombers the centerpiece of airpower spending until 1970.⁸⁸ McNamara accepted the agreement. Vinson would hold the other man to his word—on March 7, 1962, Vinson’s committee amended the National Defense Authorization Act for that year to explicitly require the Kennedy Administration to spend \$491 million on the RS-70. In a subsequent press conference, Vinson explicitly defined his action as a stand against executive overreach, and stated later in ironic understatement that while his committee didn’t “want to run the Department of Defense,” it did want to “sit at the table and get across an idea once in a while.”⁸⁹

On the same day, the Air Force Scientific Advisory Board, chaired by Gen. Jimmy Doolittle, reported to LeMay that technical obstacles would likely stand in the way of the RS-70 performing as the Air Force expected—LeMay continued his crusade for the plane regardless: on the 15th of March his Colonel David C. Jones, head of the manned systems branch of the Air Force, would testify before the House Appropriations Subcommittee on Defense in favor of the RS-70, describing it as “within the state of the art.”⁹⁰ The fight was no longer based on a practical technical belief in the plane, but rather about the preservation of the manned bomber force as an institution. Vinson held a similar position—on the 14th, McNamara briefed him privately in the Pentagon on the RS-70’s technical difficulties, but could not move the iron-willed chairman, who promised to protect his mandate.⁹¹ McNamara made his own public statement about the RS-70’s lack of technical feasibility on the 15th, but had little effect.⁹²

⁸⁶ James F. Cook, *Carl Vinson: Patriarch of the Armed Forces* (Macon, GA: Mercer University Press, 2004), 294.

⁸⁷ Ibid, 295.

⁸⁸ Matusheski, “Bomber Wars”, 21. Brown, *Flying Blind*, 224.

⁸⁹ Futrell, *Ideas, Concepts, Doctrine*, 53. Matusheski, “Bomber Wars”, 21. Kotz, *Wild Blue Yonder*, 75

⁹⁰ Nalty, *Advanced Manned Strategic Bomber* 9. Futrell 53

⁹¹ Futrell, *Ideas, Concepts, Doctrine*, 53.

⁹² Ibid, 52-53. Kotz, *Wild Blue Yonder*, 73.

The 20th of March would see the matter resolved. That morning, President Kennedy made a public declaration that the executive must give “every possible consideration” to the views of congress—Kennedy and Vinson would meet for a walk in the Rose Garden later that afternoon. The encounter is memorably recounted by several authors; James F. Cook, his biographer, provides the most detail. While walking, the President told Vinson: “Uncle Carl...”—referring to the representative by his commonly-used nickname—“this kind of language and my ignoring it will only hurt us and the country. Let me write you a letter that will get us both off this limb.” Vinson, in response, only pulled out a letter that he had already prepared, to which Kennedy reacted “That’s why you got the name ‘Swamp Fox’.” That night, McNamara and White House special counsel Ted Sorenson polished Vinson’s letter—which promised further study of the RS-70—sending the finished product back to Vinson under the seal of the President. The next morning, Vinson read the letter aloud in his committee, and subsequently moved to strike the language mandating RS-70 funding from the NDAA. While at the time Vinson insisted that the letter represented concrete progress toward getting the RS-70 built, ultimately, this was a surrender. Again in the words of his biographer “... for the good of the country, [Vinson] had acceded to the power of a commander in chief he did not wish to further embarrass.”⁹³ Vinson would retire only three years later, in 1965, as the most senior member of the House of Representatives.

While congress continued to appropriate additional funds for RS-70 development, and the Department of Defense would continue to study the aircraft, the remainder of the battle would prove perfunctory—future studies unsurprisingly concluded that the RS-70 was unnecessary, questioning whether it was “worth several billion dollars of national resources to be able to overfly Soviet targets with a few score manned bombers looking for residual weapons... after each country had exchanged several thousand megatons of nuclear fire-power on their respective target systems.”⁹⁴ The B-70—now called the XB-70, as it would never make it out of its prototyping stage—would face cost overruns due to the expensive new materials required to make compression lift technology possible⁹⁵ While LeMay would continue to advocate for the plane, his comments took on a tenor of defeat; he referred to himself as a “good soldier” who would respond to the concern of policymakers, despite his individual concerns about the fate of the bomber force.⁹⁶ In mid-1963, the house and senate both voted against additional funds for the RS-70, and LeMay, the pre-eminent advocate of the manned bomber, would throw in the towel, stating “I feel the B-70 program is dead.”⁹⁷ While the XB-70 prototypes would

⁹³ Futrell, *Ideas, Concepts, Doctrine*, 53. Brown, *Flying Blind*, 224-225, Matusheski, “Bomber Wars”, 21. Cook 296-297.

⁹⁴ Futrell, *Ideas, Concepts, Doctrine*, 55-56. Kotz, *Wild Blue Yonder*, 74.

⁹⁵ Futrell, *Ideas, Concepts, Doctrine*, 101.

⁹⁶ Ibid, 63. Nalty, *Advanced Manned Strategic Bomber*, 14.

⁹⁷ Futrell, *Ideas, Concepts, Doctrine*, 101.

eventually fly, the Air Force would not fully acquire another manned strategic bomber until the mid-1980s.

Analysis

The B-70 case shows where the military's guardianship advantage can run aground—though it is notable that it occurred in a substantially different period for executive-legislative relations. Prior to the impoundments control act of 1973, the executive had a much broader power to withhold congressionally authorized funds from the services. Nevertheless, the fact that Carl Vinson was able to force the matter to a head and demand some form of negotiation from the executive indicates that the congress did hold some power to prevent impoundments. The difference in this case from the previous, then, is that the key congressmen was effectively convinced to yield to the executive's decision. The different situations of Carl Vinson and Angus King surely played some role—King held significant power as an independent in a split congress, and while Vinson held his own form of power, he was also swayed by party loyalty.

The military's capacity to influence the legislative branch was once again evident in this case. LeMay and his allies used their testimony to convince the congress to repeatedly fund the B-70 against the executive's recommendation, regardless of the technical testimony that both they and LeMay had received indicating that the plane would face a troubled development. McNamara was unable to effectively sway congress to the same extent as key military leaders—the guardianship advantage, and the nuclear expertise that SAC and the Air Force seemed to command, likely influenced congress' alignment with them. The distinction between the Navy in the SLCM-N case and the Air Force in the B-70 case seems to be the lack of an additional targeting body outside of the service in question—STRATCOM could undermine the Navy's operational concerns, but the Air Force and SAC were able to effectively coordinate due to their shared interests.

Technologists again appear to have been beholden to military demands and constraints, rather than exerting significant influence on the decision-making process. While North American spread production between multiple states, creating a broad political base which may have influenced the broad congressional opposition to the program's cancellation, they were unable to leverage that base for a continuation of the B-70 program, and were ultimately subject to Vinson's final decision.

Section III: Analysis & Conclusion

Overall, the process of nuclear weapons acquisition seems to empower the body that does the targeting above all other parties. Congress, rather than serving as a mouthpiece for public opinion, consistently served as a representative for military interests. Both cases saw congress advocating for what the military wanted—in particular, organizations who can authoritatively articulate requirements for deterrence: the Air Force with its subordinated Strategic Air Command in the 1960s, and STRATCOM in the 2020s.

The concept of ‘flexibility’ played a key role in facilitating this empowerment of targetters. The B-70 is the first instance of an armed service explicitly leveraging a strategic theory in order to bypass executive opposition to a weapons system. In that capacity, it serves as a pioneering example of a phenomenon that would shape weapons policy into the 21st century—a phenomenon that, indeed, has appeared again in the battle over SLCM-N. In both instances, the targeting body—STRATCOM or AF/SAC—advocated for more weapons by construing their buildup as providing additional military options to the President. Targeting bodies have leveraged a socially invented mission—which may have validity militarily, but is nevertheless socially invented—to justify additional weapons acquisition.

Changes found in the transition between SAC and STRATCOM can be identified from these two cases. In particular, SAC was subordinated to the Air Force, and the two bodies advocated for each other's interests. Both were attached to high altitude bombers despite emerging studies revealing the rising threat of Soviet high-altitude air defense and corresponding requirement for low-altitude capability, so the B-70 remained primarily a high-altitude aircraft. STRATCOM's segregation from any particular service means its decisions on deterrence are no longer subject to the concerns of the services. While the Navy seems to oppose SLCM-N's development due to the operational challenges it would pose for attack submarines, STRATCOM presses on.

Targeting bodies are not even subject to the constraints of industry: Industry actors did not exert significant influence in either case—both the SLCM-N and the B-70 subordinated industrial interests to politically-stated or military-advocated requirements. It is possible that this lack of influence stemmed from, or is related to, a seeming lack of interest from congress in technical constraints—congress pushed the B-70 forward despite its technical difficulties and beyond-state-of-the-art nature, and has pushed the SLCM-N forward despite Naval concerns regarding industrial capacity and the adaptability of the Virginia-class submarine. This constraint is likely structural—congressmembers with divided attention simply cannot consider these technical factors deeply enough to give them real credence.

The targeting apparatus was only effectively restrained through the powerful confluence of executive opposition and legislative surrender—when Carl Vinson gave up the B-70. Outside of this instance, SAC and STRATCOM's powers seem extreme, as evidenced by the Eisenhower administration's reversal of its B-70 decision in response to a political uproar caused, at its root, by SAC.

What does it mean that targeters—SAC, and now STRATCOM—stand as the most powerful operators in the nuclear weapons acquisition process—at times prevailing even over the President? Who is empowered when targeters rule? One answer seems to be the software and machine learning industry. Targeting is an increasingly automated process, accomplished by advanced machine learning algorithms provided by companies such as Palantir. The increasing software-ization of targeting has empowered Palantir and companies like it.

Additionally, when targeters shape the weapons that the armed services deploy, they also exert influence on the promotion pathways and leadership structures of those services. Institutions are created within the armed services dedicated to the weapons that the targeters suggest—the officers that manage those institutions are educated in the utility of their weapons based on the justification laid out by the targeting institution. If these officers then go on to serve in higher leadership positions, they continue advocating for the principles of the targeters, effectively nudging the opinion of the service toward that of the targeting body.

The supremacy of targeting has also damaged the rigor of the traditional acquisitions process. The armed services each enjoyed acquiring exquisite and bespoke weapons systems. Since they handled these weapons day to day, and were responsible for their maintenance, they favored having high-quality resources at their disposal. Targeters care less about the quality of the munitions they employ and the platforms from which they're employed, since they're not the ones using the weapons. Rather, they separate capabilities from quality, abstracting away the ability of a weapon to strike a target from its actual technical details. This leads to situations like that of SLCM-N, where the targeter advocates for a weapon regardless of the deploying service's technical concerns. The expanding relationship of the U.S. military with commercial contractors manufacturing non-bespoke systems can be considered an outgrowth of this phenomenon.

Altogether, the military stands on rocky ground, subject to an increasingly software-ized process that has been abstracted from the experience of the warfighter. The defense policy community should work to empower the armed services and restore more methodical defense planning that takes the day-to-day

experience of equipment management into consideration, rather than continuing to skew military acquisitions in favor of targeters.

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